EBN 111/122
ELECTRICITY AND ELECTRONICS

Chapter 9
Sinusoidal Steady-State Analysis

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Sinusoids and Phasor
Chapter 9

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9.3 Phasors

- A phasor is a complex number that represents the amplitude and phase of a sinusoid (COSINE).

Complex Numbers:

$$z = x + jy$$

Complex Number — Real part — Imaginary part

Three different forms of writing:

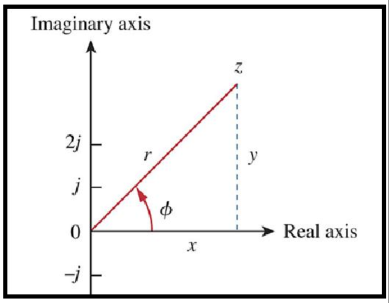
a. Rectangular: $z = x + jy = r(\cos\phi + j\sin\phi)$

b. Polar: $z = r\angle\phi°$ (r = magnitude of z)

c. Exponential: $z = re^{j\phi}$

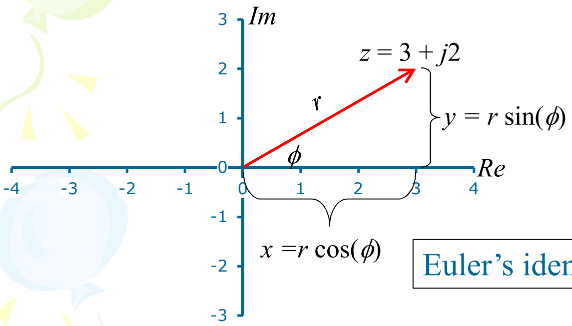
where $r = \sqrt{x^2 + y^2}$

$\phi = \tan^{-1}\dfrac{y}{x}$



9.3 Phasors

On the complex plane



$$\begin{aligned} z &= x + jy \\ &= r\cos\phi + j\,r\sin\phi \\ &= r(\cos\phi + j\,\sin\phi) \\ &= re^{\,j\phi} \\ &= r\angle\phi° \end{aligned}$$

Euler's identity: $e^{\pm j\phi} = \cos\phi \pm j\sin\phi$

A complex number consists out of:

- A Real part (x) and Imaginary part (y):

  $z = x + jy$

  OR

- A Magnitude (r) and a phase (ϕ):

  $z = r\angle\phi° = re^{\,j\phi}$

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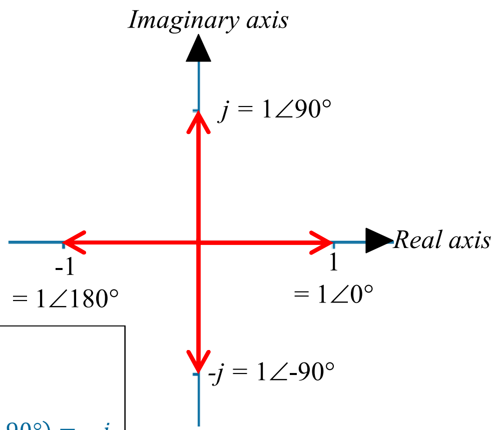
9.3 Phasors
On the complex plane



$$\sqrt{-1} = j$$

$$\frac{1}{j} = \frac{1}{1\angle 90°} = 1\angle(-90°) = -j$$

$$1 + j = \sqrt{2}\angle 45°$$

$$1 - j = \sqrt{2}\angle(-45°)$$

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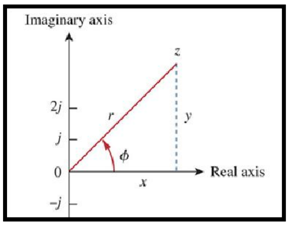
9.3 Phasors
AC signal as a Phasor

Euler's identity: $e^{\pm j\phi} = \cos\phi \pm j\sin\phi$

$$Re\left\{e^{j\phi}\right\} = \cos\phi$$

$$Im\left\{e^{j\phi}\right\} = \sin\phi$$



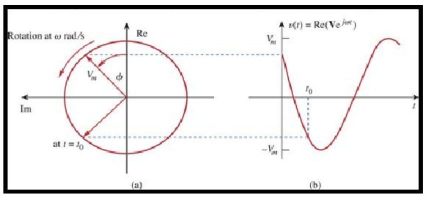
$$v(t) = V_m\cos(\omega t + \phi) = Re\left\{V_m e^{j(\omega t+\phi)}\right\}$$

$$= Re\left\{V_m e^{j\phi} e^{j\omega t}\right\}$$

$$= Re\{\boldsymbol{V} e^{j\omega t}\}$$

$$e^{j\omega t} = \cos\omega t + j\sin\omega t$$

$$\boldsymbol{V} = \bar{V} = V_m e^{j\phi} = V_m\angle\phi$$
$$= V_m\cos\phi + jV_m\sin\phi$$
$$= x + jy$$



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9.3 Phasors

Transformations

- Transform a sinusoid to and from the time domain to the phasor domain:

$$v(t) = V_m \cos(\omega t + \phi) \longleftrightarrow V = V_m \angle \phi$$

(time domain) (phasor domain)

- Amplitude and phase difference are two principal concerns in the study of voltage and current sinusoids.
- Phasor will be defined from the cosine function in all our proceeding study. If a voltage or current expression is in the form of a sine, it will be changed to a cosine by subtracting from the phase.

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9.3 Phasors

Sinusoid-phasor Transformation

Time domain representation	Phasor domain representation
$V_m \cos(\omega t + \phi)$	$V_m \underline{/\phi}$
$V_m \sin(\omega t + \phi)$	$V_m \underline{/\phi - 90^\circ}$
$I_m \cos(\omega t + \theta)$	$I_m \underline{/\theta}$
$I_m \sin(\omega t + \theta)$	$I_m /\theta - 90^\circ$
$\int v\,dt$	$\dfrac{V}{j\omega}$
$\dfrac{dv}{dt}$	$j\omega V$

$$v(t) = V_m \cos(\omega t + \phi)$$

$$\frac{dv(t)}{dt} = \frac{d}{dt} Re\left\{V_m e^{j\phi} e^{j\omega t}\right\}$$

$$= Re\left\{V_m e^{j\phi} \frac{d}{dt} e^{j\omega t}\right\}$$

$$= Re\left\{\boxed{V_m e^{j\phi} j\omega} e^{j\omega t}\right\}$$

$$\boldsymbol{V} j\omega$$

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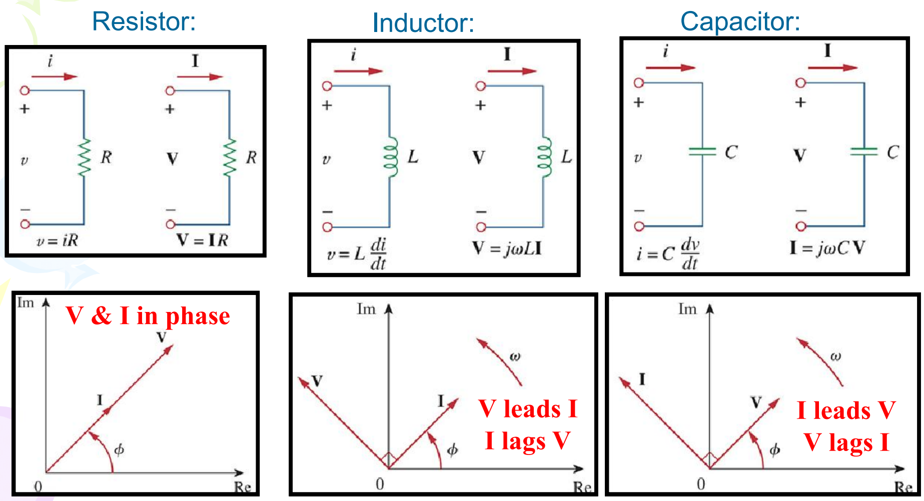
9.4 Phasors and Circuit Elements



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9.4 Phasors and Circuit Elements
Voltage-Current Relationship

Summary of voltage-current relationship		
Element	Time domain	Frequency domain
R	$v = Ri$	$V = RI$
L	$v = L\dfrac{di}{dt}$	$V = j\omega LI$
C	$i = C\dfrac{dv}{dt}$	$V = \dfrac{I}{j\omega C}$

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